MATH 161A - Applied Probability & Statistics

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1. Skipped

2. Probability

2.1 Sample Spaces and Events

Probability

- Probability is used as a mathematical tool to understand or describe random phenomena, chance variation, and uncertainty.
- We will use probability as a tool to evaluate the reliability of our conclusions about the population when we only have sample information.
- Probablility is used in a variety of fields including genetics, quality control, and finance.
- Probability is used as a model for situation for which outcomes occur randomly.
- Such situations are called **experiments**.
- Examples:
  - Toss a coin.
  - Roll a die.
  - Draw a card from a deck.

Vocabulary

- An action or process whose outcome is subject to uncertainty is called an **experiment**.
- The possible outcomes of an experiment are called **sample space**.
- An **event** is a specific outcome or a set of outcomes from an experiment.
- A **simple event** is an event that consists of a single outcome from sample space S.

Example What is the outcome of rolling an even (E), rolling greater than 3 (F), and rolling a 2 (G) on a 6-sided die?

$$S = 1, 2, 3, 4, 5, 6$$
$$E = 2, 4, 6$$
$$F = 4, 5, 6$$
$$G = 2$$

A Note on Venn Diagrams All Venn Diagrams in this course require a rectangle to represent the sample space. Events are represented by circles within the rectangle.

Operations

Set Operations

- $A \subseteq B$: A is a subset of B.
- $A \cap B$: the intersection of A and B (elements in both A and B).
- $A \cup B$: the union of A and B (elements in either A or B).
- A^c or A': the complement of A (elements in S that are not in A).

Logical Operations

- A or B: $A \vee B$
- A xor B: $A \oplus B$
- A and B: $A \wedge B$
- not A: $\neg A$

Mutually Exclusive Events

- Events A and B are **mutually exclusive** if they cannot both occur at the same time.
- In terms of set operations: $A \cap B = \emptyset$.
- Example: Rolling a die and getting an even number (E) or an odd number (O) are mutually exclusive events.

Identities

$$A \cup A' = S$$
$$A \cap A' = \emptyset$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$
$$\neg(A \vee B) = \neg A \wedge \neg B$$
$$\neg(A \wedge B) = \neg A \vee \neg B$$

2.2 Probability

Definition Probability is a function that maps the sample space S to $\mathbb{R}$. The sample space is the **domain** and the real numbers are the **codomain**.

Axioms of Probability

1. Non-negativity: For any event A, $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B, $P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i)$.

$$P(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i)$$

Not axioms but useful observations

1. $P(A) < 1$ for any event A.
2. $P(A^c) = 1 - P(A)$.
3. $P(A \cap B) = P(A) + P(B) - P(A \cup B)$ for any two events A and B.

Proving Observations

Observation 1: $P(\emptyset) = 0$

$$\begin{aligned} P(\emptyset) &= 0 \\ \emptyset \cup S &= S \\ \emptyset \cap S &= \emptyset \\ P(\emptyset \cup S) &= P(S) \\ P(\emptyset) + P(S) &= P(S) \quad \text{(by axiom 3)} \end{aligned}$$

$\emptyset$ and S are mutually exclusive.
Therefore, $P(\emptyset) = 0$ by algebra.

Observation 2: $P(A^c) = 1 - P(A)$

$$\begin{aligned}
S &= A \cup A^c \\
\emptyset &= A \cap A^c \\
P(A^c \cup A) &= P(S) \\
P(A^c) + P(A) &= P(S) \quad \text{(by axiom 3)} \\
P(A^c) + P(A) &= 1 \\
P(A^c) &= 1 - P(A)
\end{aligned}$$

Observation 3: $P(A \cup B) = P(A) + P(B)$ **if** $A \cap B = \emptyset$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad \text{for mutually exclusive } A_i$$

Let $A_1 = A$, $A_2 = B$, $A_3 = A_4 = \emptyset$

$$\begin{aligned}
P(A \cup B \cup \emptyset \cup \emptyset) &= P(A) + P(B) + 0 \\
P(A \cup B) &= P(A) + P(B) \quad \text{(by axiom 3)}
\end{aligned}$$

Identities for the next proof

$$\begin{aligned}
P(A) &= P(A \cap B') + P(A \cap B) \\
P(B) &= P(A' \cap B) + P(A \cap B) \\
P(A \cup B) &= P(A \cap B') + P(A \cap B) + P(A' \cap B) \\
1 &= P(A \cap B') + P(A \cap B) + P(A' \cap B) + P(A' \cap B')
\end{aligned}$$

Observation 4: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\begin{aligned}
P(A \cup B) &= P(A \cap B') + P(A \cap B) + P(A' \cap B) \\
&= P(A) + P(A' \cap B) \\
&= P(A) + [P(B) - P(A \cap B)] \\
&= P(A) + P(B) - P(A \cap B)
\end{aligned}$$

Other Propositions

$A \subseteq B \implies P(A) \leq P(B)$

$$\begin{aligned}
B &= (A \cap B) \cup (A' \cap B) \\
A \cap B &= A \\
B &= A \cup (A' \cap B) \\
P(B) &= P(A) + P(A' \cap B) \quad \text{(by axiom 3)} \\
P(B) &\geq P(A) \quad \text{since } P(A' \cap B) \geq 0 \text{ (by axiom 1)} \\
\text{For any } A,\ A &\subseteq S \\
P(A) &\leq P(S) \\
\therefore P(A) &< 1
\end{aligned}$$

Example If $P(C) = .46, P(D) = .47, P(C \cup D) = .62$, find the following:

$$\begin{aligned}
P(C') &= 1 - P(C) = 1 - .46 = .54 \\
P(C \cap D) &= P(C) + P(D) - P(C \cup D) = .46 + .47 - .62 = .31 \\
P(C \cap D') &= P(C) - P(C \cap D) = .46 - .31 = .15 \\
P(C' \cup D') &= 1 - P(C \cup D) = 1 - .62 = .38
\end{aligned}$$

Probability Square Trick

Example There are 5 simple events in a sample space. $S = E_1, E_2, E_3, E_4, E_5$ with probabilities p_1, p_2, p_3, p_4, p_5. Suppose $p_3 = 2p_2 = 2p_4$ and $p_2 = 2p_1 = 2p_5$. Find p_1, p_2, p_3, p_4, p_5.

$$\begin{aligned}
p_1 + p_2 + p_3 + p_4 + p_5 &= 1 \\
p_3 &= 2p_2 \\
p_3 &= 2p_4 \\
p_4 &= p_2 \\
p_2 &= 2p_1 \\
p_2 &= 2p_5 \\
p_1 &= p_5
\end{aligned}$$

Solve for the 5 unknowns with the 5 equations:

2.3 Counting Techniques

Equally Likely Outcomes

If all outcomes in a sample space are equally likely for any event A, then

$$P(A) = \frac{\text{Number of outcomes in A}}{\text{Number of outcomes in S}} = \frac{n(A)}{n(S)}$$

where $n(A)$ is the number of outcomes in event A and $n(S)$ is the number of outcomes in sample space S.

Combinatorics

Combinatorics is the branch of mathematics dealing with combinations of objects in specific sets under certain constraints. It provides tools for counting and arranging objects. Some basic combinatorial concepts include:

- **Permutations:** The number of ways to arrange a set of objects in a specific order.
- **Combinations:** The number of ways to select a subset of objects from a larger set, without regard to the order of selection.

Counting Subsets Let $\Omega = a, b, c, d, e, f, g, h, i, j$. How would we find all possible subsets of Ω?

$$Num = 2^{len(\Omega)} = 2^{10} = 1024$$

This is because each element can either be included or excluded (2 options) from a subset, leading to 2^n possible subsets for a set with n elements.

Permutations and Combinations

Permutations A permutation is an arrangement of objects in a specific order. The number of permutations of k objects chosen from n distinct objects is given by the following formula:

$$P_{k,n} = \frac{n!}{(n-k)!} = n \times (n-1) \times (n-2) \times ... \times (n-k+1)$$

Where $P_{k,n}$ is the number of permutations of n objects taken k at a time, $n!$ is the factorial of n, and $(n-k)!$ is the factorial of $(n-k)$.

Example: Suppose we want to rank 5 candidates Amy, Bai, Catherine, David, and Eduardo in preference for a position. How many different ways can we rank (1.) all of the candidates and (2.) only 2 of the candidates?

$$1. \quad P_{5,5} = \frac{5!}{5-5!} = \frac{5!}{1} = 120$$
$$2. \quad P_{5,2} = \frac{5!}{5-2!} = \frac{5!}{3!} = 20$$

Combinations A combination is a selection of objects without regard to the order of selection. The number of combinations of k objects chosen from n distinct objects is given by the following formula:

$$C_{k,n} = \frac{n!}{k!(n-k)!} = \frac{P_{k,n}}{k!}$$

Where $C_{k,n}$ is the number of combinations of n objects taken k at a time, $n!$ is the factorial of n, $k!$ is the factorial of k, and $(n-k)!$ is the factorial of $(n-k)$.

Example: How many hands of 3 cards can be drawn from a standard deck of 52 playing cards?

$$C_{3,52} = \binom{52}{3} = \frac{52!}{3!(52-3)!} = \frac{52!}{3!49!} = 22100$$

Example: A playlist contains 100 songs. 10 of these songs are by The Strokes. We can find the probability of the first song by The Strokes being the 5th song played by finding the ratio of the number of favorable outcomes to the total number of outcomes.

$$
\begin{aligned}
P(\text{1st Strokes on 5th}) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\
&= \frac{P_{4,90} \cdot P_{1,10}}{P_{5,100}} = \frac{\frac{90!}{(90-4)!} \cdot 10}{\frac{100!}{(100-5)!}} \\
&= .0679
\end{aligned}
$$

Another way of thinking about this would be that there are $\binom{100}{10}$ different ways to choose the 10 Strokes songs. Now if we choose 9 of the last 95 songs to be Strokes songs, there are $\binom{95}{9}$ ways to do this, leaving 4 non-Strokes songs and 1 Strokes song in the first 5 songs. The probability is then:

$$
P(\text{1st Strokes on 5th}) = \frac{\binom{95}{9}}{\binom{100}{10}} = .0679
$$

Example: A university warehouse has recieved 25 printers, of which 10 are laser printers and 15 are inkjet printers. If 6 printers are randomly selected for testing, what is the probability that exacty 3 of those selected are laser printers?

Let D_3 be the event that exactly 3 of the selected printers are laser printers.

$$
\begin{aligned}
P(D_3) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\
&= \frac{N(D_3)}{N} = \frac{\binom{15}{3} \cdot \binom{10}{3}}{\binom{25}{6}} = .3083
\end{aligned}
$$

2.4 Conditional Probability

A conditional probability of an event is the probability that the event occurs given that some other event has already occurred. Conditional probabilities can prove to be counterintuitive and may lead tableofcontents suprising results.

Definition: The conditional probability of an event A given that an event B has occurred is denoted by $P(A|B)$ and is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example: If you flip 3 coins, what is the probability that exactly 2 are heads (B)?

$$\begin{aligned}
&\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\} \\
&B = \{HHT, HTH, THH\} \\
&A = \{HHH, HTT, HTH, HHT\}
\end{aligned}$$

$$\begin{aligned}
P(A \cap B) &= \frac{2}{8} \\
P(B) &= \frac{3}{8} \\
P(A|B) = \frac{P(A \cap B)}{P(B)} &= \frac{\frac{2}{8}}{\frac{3}{8}} = \frac{2}{3}
\end{aligned}$$

Multiplication Rule

The multiplication rule is a way to find the probability of the intersection of two events. It states that:

$$\begin{aligned}
P(A \cap B) = P(A|B) \cdot P(B) \\
P(A \cap B) = P(B|A) \cdot P(A)
\end{aligned}$$

CHECK THIS FOR ACCURACY

General Multiplication Rule: For any number of events $A_1, A_2, \ldots, A_n$:

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1})$$

Law of Total Probability

For any events A and B, we can get the following from the multiplication rule:

$$P(A \cap B) = P(A|B)P(B)$$
$$P(A \cap B') = P(A|B')P(B')$$

: Note that B and B' form a *partition* of Ω.

General Law of Total Probability: Let $B_1, \ldots, B_n$ be a partition of the sample space Ω. Then for any event A:

$$P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

Example: Suppose that we have 2 boxes with marbles in them. Box 1 has 1 white and 3 black marbles. Box 2 has 2 white and 2 black marbles. We randomly select a box and then randomly select a marble from that box. What is the probability that we select a black marble?

$$\begin{aligned} P(Black) &= P(Black|Box1)P(Box1) + P(Black|Box2)P(Box2) \\ &= \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) + \left(\frac{2}{4}\right)\left(\frac{1}{2}\right) \\ &= \frac{3}{8} + \frac{2}{8} = \frac{5}{8} \end{aligned}$$

Example: On any given morning, it is either raining (25% of the time), foggy (35% of the time), or sunny (40% of the time). If it is raining, there is a 30% change that I will miss the bus. If it is foggy, there is a 20% chance that I will miss the bus. If it is sunny, there is a 10% chance that I will miss the bus. What is the probability that I will miss the bus on any given morning?

$$\begin{aligned} P(MB) &= P(MB|R)P(R) + P(MB|F)P(F) + P(MB|S)P(S) \\ &= (0.30)(0.25) + (0.20)(0.35) + (0.10)(0.40) \\ &= 0.075 + 0.07 + 0.04 = 0.185 \end{aligned}$$

Bayesian Analysis

Bayesian analysis is a statistical method that involves using Bayes' theorem to update the probability of a hypothesis based on new evidence or data. It is named after Thomas Bayes, an 18th-century statistician and theologian.

Bayes' Theorem Let $B_1, \ldots, B_n$ form a partition of the sample space Ω. Then for any event A:

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{n} P(A|B_j)P(B_j)}$$

Note that in the special case of the partition consisting of B and B', this reduces to:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')}$$

Example: If I pull a black marble from one of the boxes in the previous example, what is the probability that I pulled it from Box 1?

$$\begin{aligned} P(B_1|Black) &= \frac{P(Black|B_1)P(B_1)}{P(Black)} \\ &= \frac{(\frac{3}{4})(\frac{1}{2})}{\frac{5}{8}} = \frac{3}{5} \end{aligned}$$

Example: According to the Arizona Chapter of the American Lung Association, 7% of the population has lung disease. Of those, people, 90% are smokers; and of those without lung disease, 74.7% are non-smokers. What are the chances that a smoker has lung disease?

$$\begin{aligned}
P(LD|S) &= \frac{P(S|LD)P(LD)}{P(S|LD)P(LD) + P(S|LD')P(LD')} \\
&= \frac{(0.90)(0.07)}{(0.90)(0.07) + (0.253)(0.93)} \\
&= \frac{0.063}{0.063 + 0.23529} = \frac{0.063}{0.29829} \approx 0.2111
\end{aligned}$$

Example: You give a friend a letter to mail. He forgets to mail it with probability 0.2. Given that he mails it, the Post Office delivers it with probability 0.9. Given that the letter was not delivered, what is the probability that it was not mailed.

$$\begin{aligned}
P(M'|D') &= \frac{P(D'|M')P(M')}{P(D'|M')P(M') + P(D'|M)P(M)} \\
&= \frac{(1)(0.2)}{(1)(0.2) + (0.1)(0.8)} \\
&= \frac{0.2}{0.2 + 0.08} = \frac{0.2}{0.28} = \frac{5}{7}
\end{aligned}$$

2.5 Independence

Definition: Two events A and B are independent if the occurance of one event does not affect the probability of the other event occurring. Otherwise, the events are dependent. Sometimes it is obvios that two events influence each other, such as crude oil prices and gas prices. Sometimes it is obvious that two events are not dependent, such as rolling a die and the weather. Sometimes it is difficult to tell, such as obtaining two heads in three coin flips and obtaining a head on the first of three flips. Mutual exclusivity and independence are not the same thing. Two events can be independent and not mutually exclusive, such as rolling a 1 and rolling an odd number on a die.

Checking for Independence: In many cases, you will need to evaluate independence by checking the following formulas:

- A and B are independent if and only if $P(A|B) = P(A)$
- A and B are independent if and only if $P(A \cap B) = P(A)P(B)$

Note that independent trials are okay to be assumed to be independent, such as rolling a die or flipping a coin.

Properties:

- If A and B are independent, then b and A are independent.
- If A and B are independent, then A' and B', A and B', and A' and B are all independent.

Example: Suppose we draw a card from a deck of 52 playing cards. Let A be the event that the card is an Ace, and let B be the event that the card is a Spade. Are A and B independent?

$$\begin{aligned}
P(A) &= \frac{4}{52} = \frac{1}{13} \\
P(B) &= \frac{13}{52} = \frac{1}{4} \\
P(A \cap B) &= \frac{1}{52} \\
P(A)P(B) &= \frac{1}{13} \cdot \frac{1}{4} = \frac{1}{52}
\end{aligned}$$

Since $P(A \cap B) = P(A)P(B)$, A and B are independent.

Example: Suppose we roll 2d6. What is the probability of getting 2 even numbers?

$$\begin{aligned}
P(E_1 \cap E_2) &= P(E_1)P(E_2) && \text{since } E_1 \text{ and } E_2 \text{ are independent} \\
&= \left(\frac{3}{6}\right)\left(\frac{3}{6}\right) \\
&= \frac{9}{36} = \frac{1}{4}
\end{aligned}$$

Example: Roll 2d6. Let A be the event that the sum is 7. Let B be the event that the first die is a 4. Are A and B independent?

$$\begin{aligned}
P(A) &= \frac{6}{36} = \frac{1}{6} \\
P(B) &= \frac{1}{6} \\
P(A \cap B) &= \frac{1}{36} \\
P(A)P(B) &= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}
\end{aligned}$$

Since $P(A \cap B) = P(A)P(B)$, A and B are independent.

3. Discrete Random Variables and Probability Distributions

3.1 Random Variables

In many random experiments, we are not interested in the specific outcome of an experiment, but in some characteristic of that outcome. If you play the lottery , for example, you would like the numbers chosen in the weekly drawing to match as many of the number you marked on your ticket as possible. You do not particularly care which of your numbers are matched, but rather how many matches you have.

Definition For a given sample space of some experiment, a **random variable** is any rule that associates a number with each outcome in S. In mathematical language, a random variable is a function whose domain is the sample space S and whose range is the set of real numbers. We usually denote random variables with capital letters such as X, Y, and Z.

$$X : S \to \mathbb{R}$$
$$\exists s \in S \ni X(s) = x$$

Random Variable Examples

- Sum of two dice
- Number of defects on a randomly selected piece of fabric.
- SAT score of a randomly selected student.
- Number of heads in 10 flips of a coin.
- The amount a company has to pay to an employee in pension.

Types of Random Variables

- A **discrete random variable** is a random variable that can take on a finite or countably infinite number of possible values.
- A **continuous random variable** is a random variable that can take on infinitely many values corresponding to an interval on the real line.

3.2 Probability Distributions of Discrete Random Variables

Probability Distributions

Definition To describe a discrete random variable X, you need two pieces of information:

- What values X can assume.
- The probabilities associated with each of these values.

The **probability distribution** or **probability mass function** (pmf) of a discrete random variable X is a formula, table, or graph that gives both the possible values of X and the probabilities associated with each value of X. The probability distribution or pmf of a discrete random variable is defined for every number x by:

$$p(x) = P(X = x)P(\forall s \in S, X(s) = x)$$

Example: Let X be the random variable that gives the sum of 2d6.

x	2	3	4	5	6	7	8	9	10	11	12
p(x)	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Bernoulli Random Variables

Definition Any random variable whose only possible values are 0 and 1 is called a **Bernoulli random variable**. The PMF looks like this:

x	0	1
p(x)	$1-p$	p

Example: A box contains one red, two orange, and five black balls. You choose two balls at random. For each red ball you win \$5, for each orange ball you win \$1, and for each black ball you win nothing. Let X be the random variable that gives your total winnings. Find the probability distribution of X and the probability that you will win at most \$3.

x	p(x)	Explanation
0	$\frac{10}{28}$	BB
1	$\frac{10}{28}$	BO, OB
2	$\frac{1}{28}$	OO
5	$\frac{5}{28}$	RB, BR
6	$\frac{2}{28}$	RO, OR

$$P(X \leq 3) = p(0) + p(1) + p(2) = \frac{10}{28} + \frac{10}{28} + \frac{1}{28} = \frac{21}{28} = \frac{3}{4}$$

Cumulative Distribution Function

Definition The **cumulative distribution function** (CDF) $F(x)$ of a discrete RV X with PMF $p(x)$ is defined for **every number x** by:

$$F(x) = P(X \leq x) = \sum_{i \leq x} p(i)$$

Example: Take the previous example with the balls. Some examples of using CDF here are:

$$\begin{aligned}
F(3) &= P(X \leq 3) = p(0) + p(1) + p(2) = \frac{21}{28} = \frac{3}{4} \\
F(5) &= P(X \leq 5) = p(0) + p(1) + p(2) + p(5) = \frac{26}{28} = \frac{13}{14} \\
F(-10) &= P(X \leq -10) = 0 \\
F(10) &= P(X \leq 10) = 1
\end{aligned}$$

We can find the CDF:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{10}{28} & 0 \leq x < 1 \\ \frac{20}{28} & 1 \leq x < 2 \\ \frac{21}{28} & 2 \leq x < 5 \\ \frac{26}{28} & 5 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

Example: Find the PMF from the following CDF:

$$\begin{cases} 0 & x < -3 \\ 0.2 & -3 \leq x < -2 \\ 0.3 & -2 \leq x < 0 \\ 0.6 & 0 \leq x < 1 \\ 0.9 & 1 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

PMF:

x	p(x)	Explanation
-3	0.2	$F(-3) - F(-\infty)$
-2	0.1	$F(-2) - F(-3)$
0	0.3	$F(0) - F(-2)$
1	0.3	$F(1) - F(0)$
4	0.1	$F(4) - F(1)$

3.3 Expected Values

The **expected value** is a weighted average of the possible values of a random variable X with PMF p(x). It is given by:

$$E(X) = \mu_X = \sum_x xp(x)$$

where the sum is over all possible values of X.

Names for Expected Value

- Mean
- Expected Value
- Expectation
- μ
- $E(X)$
- (Bonus) The "center of mass" of the histogram of the PMF.

Example: Craps is a dice game in which the players make wagers on the sum of a roll of a pair of dice. One of the wagers available to the player is the *field bet*. The field consists of $\{5, 6, 7, 8\}$. If the sum of the dice is one of these numbers, you lose your bet. Otherwise you get your bet back plus a payout according to the following table:

Winnings	-1	1	2	3
Outcome	field	3, 4, 9, 10, 11	2	12

Is the bet favorable to the player?

Solution: Let X be the random variable that gives your winnings on a field bet. The PMF is given by:

x	-1	1	2	3
$p(x)$	$\frac{16}{36}$	$\frac{20}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

$$\begin{aligned} E(X) &= \sum_x xp(x) \\ &= -1 \cdot \frac{20}{36} + 1 \cdot \frac{14}{36} + 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{36} \\ &= -\frac{20}{36} + \frac{14}{36} + \frac{2}{36} + \frac{3}{36} \\ &= -\frac{1}{36} \\ &\approx -0.028 \end{aligned}$$

The expected value is negative, so the bet is unfavorable to the player.

Expectation of a Function of a Random Variable

If X is a discrete random variable with PMF $p(X)$ and $h(x)$ is any real valued function of X, then the expectation of $h(X)$ is given by:

$$E[h(X)] = \sum_x h(x)p(x)$$

Example: Find $E(X^2)$ if X is the sum of two dice.

$$E(X^2) = \sum x^2 p(x)$$

Note that this is not always reversible. $E[h(X)] \neq h(E[X])$ except when $h(x)$ is a linear function. Then

$$E(aX + b) = aE[X] + b$$

Proof:

$$\begin{aligned} E(aX + b) &= \sum_x (ax + b)p(X) \\ &= \sum_x ax \cdot p(x) + b \cdot p(x) \\ &= a \sum_x xp(x) + b \sum_x p(x) \\ &= aE[X] + b \cdot 1 \\ &= aE[X] + b \end{aligned}$$

Special Cases and Rules:

$$\begin{aligned} a = 0 &: E(b) = b \\ b = 0 &: E(aX) = aE[X] \\ E(X + Y) &= E(X) + E(Y) \\ E(3X - 2Y + b) &= 3E(X) - 2E(Y) + b \end{aligned}$$

Variance of a Discrete Random Variable

The **variance** of a discrete random variable is the "spread" of the random variable around it's mean. Suppose X is a discrete random variable with PMF p(x) and mean μ. The variance of X, denoted by σ^2 is given by:

$$\sigma^2 = Var(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$$

Recall here that $\mu = E(X)$. The variance is the mean squared distance of the values of X from their mean.

Names for Variance

- Variance
- $Var(X)$
- $V(X)$
- σ^2

Computation Formula

$$V(X) = E[X^2] - (E[X])^2$$

Standard Deviation

The **standard deviation** of a discrete random variable X is the positive square root of the variance of X. It is denoted by σ.

$$\sigma = SD(X) = \sqrt{Var(X)} = \sqrt{E[X^2] - (E[X])^2}$$

Note: Variance is NOT a linear operator. Let X be a random variable with constants a and b. Then:

$$V(aX + b) = a^2 V(X) \implies \sigma_{aX+b} = |a| \sigma_X$$

Example: Show that the variance of the Bernoulli Random Variable is $p(1-p)$:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \sum_x x^2 p(x) - \left(\sum_x x p(x) \right)^2 \\ &= 0^2(1-p) + 1^2(p) - (0(1-p) + 1(p))^2 \\ &= p - p^2 \\ &= p(1-p) \end{aligned}$$

Example: Let X be a discrete random variable with $\mu = 2$ and variance of $\sigma^2 = 6$. (a) Find $E(X^2)$. (b) Find $V[(X-1)^2]$ and (c) Find σ_X.

$$
\begin{aligned}
(a) \quad E(X^2) &= V(X) + [E(X)]^2 \\
&= 6 + 2^2 \\
&= 10
\end{aligned}
$$

$$
\begin{aligned}
(b) \quad V[(X-1)^2] &= E[(X-1)^4] - [E(X-1)^2]^2 \\
&= E[X^4 - 4X^3 + 6X^2 - 4X + 1] - [E(X^2) - 2E(X) + 1]^2 \\
&= E[X^4] - 4E[X^3] + 6E[X^2] - 4E[X] + 1 - [10 - 4 + 1]^2 \\
&= E[X^4] - 4E[X^3] + 60 - 8 + 1 - 49 \\
&= E[X^4] - 4E[X^3] + 4
\end{aligned}
$$

$$
(c) \quad \sigma_X = \sqrt{V(X)} = \sqrt{6} \approx 2.45
$$

3.4 The Binomial Probability Distribution

Definition: There are many experiments that conform either exactly or approximately to the following list of requirements:

- The experiment consists of a sequence of n identical trials.
- Each trial results in one of two outcomes, which we will call a success and a failure.
- The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
- The probability of a success, denoted by p, remains the same from trial to trial. The probability of a failure is $q = 1 - p$.

An experiment that satisfies these requirements is called a **binomial experiment**.

Example: Consider each of the next n vehicles undergoing an emissions test, and let S denote a cehicle thta passes the test and F denote one that fails.

- The experiment consists of n identical trials, where each trial is the emissions test of a vehicle.
- Each trial results in one of two outcomes: S or F.
- The trials are independent, assuming that the vehicles are randomly selected.
- The probability of a success (a vehicle passing the test) remains the same from trial to trial, assuming that the vehicles are randomly selected.

Thus, this is a binomial experiment.

The Binomail Random Variable and Distribution

In most binomial experiments, we are interested in the total number of successes that occur in the n trials rather than the specific sequence of successes and failures.

Definition: The **binomial random variable** X associated with a binomial experiment consisting of n trials is defined as:

$$X = \text{the number of successes in } n \text{ trials.}$$

Example: Suppose that $n = 3$. There are 8 possible outcomes of the experiment, as shown in the following table:

Outcome	X
SSS	3
SSF	2
SFS	2
FSS	2
SFF	1
FSF	1
FFS	1
FFF	0

The possible values of X are 0, 1, 2, and 3.

Example: Consider the case that $n = 4$. Find P(SSFS) and P(SSSS):

$$\begin{aligned} P(SSFS) &= P(S) \cdot P(S) \cdot P(F) \cdot P(S) \\ &= p \cdot p \cdot (1-p) \cdot p \\ &= p^3(1-p) \\ P(SSSS) &= P(S) \cdot P(S) \cdot P(S) \cdot P(S) \\ &= p \cdot p \cdot p \cdot p \\ &= p^4 \end{aligned}$$

Notation: Because the pmf of a binomial random variable X depends on the two parameters n and p, we denote the pmf by $b(x : n, p)$.

Example: Find $b(3; 4, p)$:

$$\begin{aligned} b(3;4,p) &= P(X=3) \\ &= P(SSSF) + P(SSFS) + P(SFSS) + P(FSSS) \\ &= p^3(1-p) + p^3(1-p) + p^3(1-p) + p^3(1-p) \\ &= 4p^3(1-p) \end{aligned}$$

Theorem :

$$b(x;n,p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

Example: Each of six randomly selected cola drinkers is given a glass containing cola S and one containing cola F. The glasses are identical in appearance except for a code on the bottom to identify the cola. Suppose there is actually no tendency among cola drinkers to prefer one cola over the other. Then $p = P(S) = 0.5$. Find (1) the probability that exactly four of the six cola drinkers prefer cola S, (2) the probability that at least four of the six cola drinkers prefer cola S, and (3) the probability that at most 1 of the six cola drinkers prefer cola S.

1.

$$\begin{aligned} P(X=4) = b(4;6,0.5) &= \binom{6}{4}(0.5)^4(1-0.5)^{6-4} \\ &= \binom{6}{4}(0.5)^4(0.5)^2 \\ &= 15 \cdot (0.5)^6 \\ &= \frac{15}{64} \approx 0.2344 \end{aligned}$$

2.

$$\begin{aligned} P(X \geq 4) &= \sum_{x=4}^{6} b(x; 6, 0.5) \\ &= \sum_{x=4}^{6} \binom{6}{x} (0.5)^x (1-0.5)^{6-x} \\ &= \frac{22}{64} \approx 0.3438 \end{aligned}$$

3.

$$\begin{aligned} P(X \leq 1) &= \sum_{x=0}^{1} b(x; 6, 0.5) \\ &= \sum_{x=0}^{1} \binom{6}{x} (0.5)^x (1-0.5)^{6-x} \\ &= \frac{7}{64} \approx 0.1094 \end{aligned}$$

Binomial Tables

For $X \sim \text{Binomial}(n, p)$, the cdf will be denoted by:

$$B(x; n, p) = P(X \leq x) = \sum_{y=0}^{\lfloor x \rfloor} b(y; n, p)$$

Theorem: If $X \sim B(n, p)$, then:

- $E(X) = np$
- $V(X) = np(1-p)$
- $\sigma_X = \sqrt{np(1-p)}$

3.5 Hypergeometric and Negative Binomial Probability Distributions

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Negative Binomial Distribution

Definition: A negative binomial experiment is a sequence of independent and identical Bernoulli trials that continues until the r^{th} success is observed. The negative binomial random variable X is defined as the number of trials required to obtain r successes, where r is a positive integer. This is denoted:

$$X \sim \text{Negative Binomial}(r,p) \equiv X \sim NB(r,p)$$

Examples:

- The number of times you flip a fair coin until you observe heads for the fourth time.
- The number of flies you catch until you have caught 10 that are female.
- The number of customers that enter a bank until 5 of them make a deposit.

Example: Suppose X = the number of times you have to flip a coin until you obtain the three heads. Find $P(X=5)$.

$$\begin{aligned} P(X=5) &= \binom{5-1}{3-1}\left(\frac{1}{2}\right)^3\left(1-\frac{1}{2}\right)^{5-3} \\ &= \binom{4}{2}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 \\ &= 6\left(\frac{1}{2}\right)^5 = \frac{6}{32} = \frac{3}{16} \end{aligned}$$

PMF: The possible values of a Negative Binomial random variable X are $x = r, r+1, r+2, ...$ The probability that the r^{th} success occurs on the k^{th} trial is given by the pmf:

$$P(X=k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, ...$$

Note: If $X_1, \ldots, X_r$ are independent geometric random variables with parameter p, then:

$$X = \sum_{i=1}^{r} X_i \sim NB(r, p)$$

Mean and Variance: The expected value and variance of a Negative Binomial random variable with parameters r and p are given by:

$$E(X) = \frac{r}{p}$$
$$V(X) = \frac{r(1-p)}{p^2}$$

Example: You roll a fair six-sided die until you see the third six. Define the random variable you are working with in words and state the corresponding distribution along with all relevant parameters. (a) What is the probability that this happens on the tenth roll? (b) How many rolls do you expect to make?

Let $X \sim NB(3, \frac{1}{6})$ be the number of rolls until the third six.

a.

$$\begin{aligned}
P(X = 10) &= \binom{10-1}{3-1}\left(\frac{1}{6}\right)^3\left(1-\frac{1}{6}\right)^{10-3} \\
&= \binom{9}{2}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^7 \\
&= 36\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^7 \\
&= 36 \cdot \frac{1}{216} \cdot \frac{78125}{279936} \\
&= \frac{36 \cdot 78125}{216 \cdot 279936} \\
&\approx 0.0845
\end{aligned}$$

b.

$$E(X) = \frac{r}{p} = \frac{3}{\frac{1}{6}} = 18$$

Geometric Probability Distribution

Definition: This is a special case of the negative binomial distribution where $r = 1$. Essentially we are looking for the number of trials required to obtain the first success. This is denoted:

$$X \sim \text{Geometric}(p)$$

The pmf is given by:

$$P(X = k) = p(1-p)^{k-1}, \quad k = 1, 2, 3, \ldots$$

The mean and variance are given by:

$$E(X) = \frac{1}{p}, \qquad V(X) = \frac{1-p}{p^2}$$

Review of Geometric Series:

$$1 + A + A^2 + A^3 + ... + A^n = \begin{cases} \frac{1-A^{n+1}}{1-A}, & A \neq 1 \\ n+1, & A = 1 \end{cases} \frac{1-A^{n+1}}{1-A}$$

$$\sum_{k=0}^{\infty} A^k = \frac{1}{1-A}, \quad |A| < 1$$

Example: You roll a fair die until you see the first six. Define the random variable you are working with in words and state the corresponding distribution along with all relevant parameters. (a) How many rolls do you expect to make? (b) What is the probability that the first six occurs on the fifth roll? (c) What is the probability that you will need more than 10 rolls?

Let $X \sim \text{Geo}(\frac{1}{6})$ be the number of rolls until the first six.

a.

$$E(X) = \frac{1}{p} = \frac{1}{\frac{1}{6}} = 6$$

b.

$$\begin{aligned} P(X=5) &= p(1-p)^{5-1} \\ &= \frac{1}{6}\left(1-\frac{1}{6}\right)^4 \\ &= \frac{1}{6}\left(\frac{5}{6}\right)^4 \\ &= \frac{625}{7776} \approx 0.0804 \end{aligned}$$

c.

$$\begin{aligned} P(X>10) &= 1 - P(X \leq 10) \\ &= 1 - \sum_{k=1}^{10} P(X=k) \\ &= 1 - \sum_{k=1}^{10} p(1-p)^{k-1} \\ &= 1 - p\sum_{k=0}^{9}(1-p)^k \\ &= 1 - p\left(\frac{1-(1-p)^{10}}{1-(1-p)}\right) = 1 - (1-(1-p)^{10}) \\ &= (1-p)^{10} = \left(1-\frac{1}{6}\right)^{10} = \left(\frac{5}{6}\right)^{10} \approx 0.1615 \end{aligned}$$

Remark: The textbook uses Y, the number of faliures before the first success, instead of X, the number of trials until the first success. This means that $Y = X - 1$.

Hypergeometric Probability Distribution

Definition: Suppose you have a population of size N, in which each individual can be classified as either a success (S) or a failure(F). The number of successes in the population is M. A sample of size n is selected at random without replacement from the population. Let X be the number of successes in the sample. Then X is a hypergeometric random variable with parameters n, M, and N:

$$X \sim \text{Hypergeometric}(n, M, N) \equiv X \sim HG(n, M, N)$$

The pmf is denoted:

$$h(n, M, N)$$

Example: Identify the parameters n, M, and N for the following situations:

- The number of aces in a poker hand.
  - $N = 52$ (cards in deck)
  - $M = 4$ (aces in deck)
  - $n = 5$ (cards in hand)
- The number of juniors in a 5-person committee chosen at random from a group of ten juniors and five seniors.
  - $N = 15$ (total students)
  - $M = 10$ (juniors)
  - $n = 5$ (committee members)
- The number of green marbles you pull when randomly selecing 3 marbles from a beg with 10 red, 8 green, and 2 blue marbles.
  - $N = 20$ (total marbles)
  - $M = 8$ (green marbles)
  - $n = 3$ (marbles selected)

Let X = the number of green marbles you pull when randomly selecting 3 marbles from a bag with 10 red, eight green, and two blue marbles. Find the PMF of X.

$$P(X=0)=\frac{\binom{8}{0}\binom{12}{3}}{\binom{20}{3}}=\frac{220}{1140}=\frac{22}{114}\approx 0.19298$$
$$P(X=1)=\frac{\binom{8}{1}\binom{12}{2}}{\binom{20}{3}}=\frac{8\cdot 66}{1140}=\frac{528}{1140}=\frac{44}{95}\approx 0.46316$$
$$P(X=2)=\frac{\binom{8}{2}\binom{12}{1}}{\binom{20}{3}}=\frac{28\cdot 12}{1140}=\frac{336}{1140}=\frac{28}{95}\approx 0.29474$$
$$P(X=3)=\frac{\binom{8}{3}\binom{12}{0}}{\binom{20}{3}}=\frac{56\cdot 1}{1140}=\frac{56}{1140}=\frac{14}{285}\approx 0.04912$$

PMF: Generically, the pmf is given by:

$$h(x;n,M,N)=P(X=x)=\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}},\quad \max\{n-(N-M),0\}\le x\le\min\{n,M\}$$

$$E(X)=n\left(\frac{M}{N}\right)$$
$$V(X)=n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Example: Let X = nuymber of aces in a 5 card poker hand. (a) Describe the distribution of X. (b) What is the probability that $x=2$? (c) How many aces do you expect to have?

a.

$$X\sim HG(5,4,52)$$

b.

$$\begin{aligned}P(X=2)&=h(2;5,4,52)\\&=\frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}\\&=\frac{6\cdot 17296}{2598960}\\&=\frac{103776}{2598960}\approx 0.03993\end{aligned}$$

c.

$$E(X) = 5\left(\frac{4}{52}\right) = \frac{5}{13} \approx 0.3846$$

Relationship Between the Binomial and Hypergeometric Distributions

Remark: The hypergeometric distribution can be approximated by the binomial distribution when the population size N and the number of successes M is large relative to the sample size n.

3.6 Poisson Probability Distribution

Definition: The **Poisson distribution** is used to model the number of times a rare event occurs.
Examples:

- The number of earthquakes in CA in a month.
- The number of misprints on a page of some document.
- The number of atoms emitted by a radioactive particle.

A random variable X is called a Poisson (λ) random variable $(\lambda > 0)$, if X takes on values $0, 1, 2, \ldots$ and has pmf:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

- The expected value $E(X) = \lambda$
- The variance $V(X) = \lambda$
- The parameter λ is the average number of occurrences in a given interval.

Note: Informally, the smaller λ is, the more rare the event X is.

Example: Suppose on average there are two typos per page in the text book.

- For a randomly selected page of the book, what is the probability that there are no typos?
- For a randomly selected page of the book, what is the probability that there are two typos?
- For a randomly selected page of the book, what is the probability that there is at least 1 typo?

Let $X \sim \text{Poisson}(2)$ be the number of typos on a randomly selected page.

- $\lambda = 2$
- $P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2} \approx 0.1353$
- $P(X = 2) = \frac{e^{-2}2^2}{2!} = 2e^{-2} \approx 0.2707$
- $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-2} \approx 0.8647$

Example: Both Hurricane Harvey and Hurricane Irma were said to be "One in 500 Year" storms, yet they both occurred in 2017.

1. What is the probability that 2 such storms occur in a given year? Same month?
2. That is the probability that a baby born today in a hurricane zone will see one such storm. What about a "one in 100 year" storm?

Let $X \sim \text{Poisson}(\lambda)$ be the number of "one in 500 year" storms in a year and $Y \sim \text{Poisson}(\mu)$ be the number of "one in 500 year" storms in a month.

1.

$$\begin{aligned}
\lambda &= \frac{1}{500} \\
P(X = 2) &= \frac{e^{-\frac{1}{500}}\left(\frac{1}{500}\right)^2}{2!} \\
&\approx 1.996 \times 10^{-6}
\end{aligned}$$

$$\begin{aligned}
\mu &= \frac{1}{500 \cdot 12} = \frac{1}{6000} \\
P(Y = 2) &= \frac{e^{-\frac{1}{6000}}\left(\frac{1}{6000}\right)^2}{2!}
\end{aligned}$$

2.

$$\begin{aligned}
\lambda &= \frac{1}{100} \\
P(X \geq 1) &= 1 - P(X = 0) = 1 - e^{-\frac{1}{100}} \\
&\approx 0.00995
\end{aligned}$$

$$\begin{aligned}
\mu &= \frac{1}{100 \cdot 12} = \frac{1}{1200} \\
P(Y \geq 1) &= 1 - P(Y = 0) = 1 - e^{-\frac{1}{1200}} \\
&\approx 0.000833
\end{aligned}$$

4. Continuous Random Variables and Probability Distributions

4.1 Probability Density Functions

Continuous Random Variables

Definition: A random variable X is said to be **continuous** if its set of possible values is an interval or a collection of intervals on the real line. Examples of continuous random variables include:

- The time required to complete a task.
- The height of a randomly selected person.
- The amount of rainfall in a given month.
- The weight of a randomly selected object.

Probability Density Function (PDF)

Definition: Let X be a continuous random variable. Then a probability distrubution or probability density function (PDF) of X is a function $f(x)$ such that for any two

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Thus the probability that X takes on a value between a and b is represented by the area under the curve of $f(x)$ from a to b.

Properties of PDF:

- $f(x) \geq 0$ for all x.
- $\int_{-\infty}^{\infty} f(x)dx = 1$.
- $P(X = x) = 0$ for any specific value of x.
- $P(X < x) = P(X \leq x)$.

Calculus Concepts Needed:

- Differentiation and Integration
- Polynomial and Exponential Functions
- Possibly Product and Quotient Rules
- Chain Rule
- Integration by Parts
- L'hopital's Rule
- $\int_0^\infty f(x)dx = \lim_{b\to\infty} \int_0^b f(x)dx$

Example: Let X be a continuous random variable with PDF:

$$f(x) = \begin{cases} cx & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

a. Find the value of c.
b. Compute $P(2 \leq X \leq 4)$

a.

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f(x)dx \\ 1 &= \int_0^3 cxdx \\ 1 &= c\left[\frac{x^2}{2}\right]_0^3 \\ 1 &= c \cdot \frac{9}{2} \\ c &= \frac{2}{9} \end{aligned}$$

b.

$$
\begin{aligned}
P(2 \leq X \leq 4) &= \int_2^4 f(x)dx \\
&= \int_2^3 \frac{2}{9}xdx + \int_3^4 0dx \qquad \text{(since } f(x) = 0 \text{ for } x > 3\text{)} \\
&= \frac{2}{9}\left[\frac{x^2}{2}\right]_2^3 + 0 \\
&= \frac{2}{9}\left(\frac{9}{2} - 2\right) \\
&= \frac{5}{9}
\end{aligned}
$$

The Uniform Distribution

In the beginning of this course we considered smaple spaces in which every outcome is equally likely. For example, let $X =$ the value of a fair six-sided die. This is called the uniform distribution. The analogous continuous distribution is called the continuous uniform distribution. We have to consider all possible values in an interval (a, b). The probability that X falls into a subinterval should only depend on the length (and not the location) of the subinterval (a, b). This means that the probability density function must be constant. Let $f(x) = c \forall x \in (a, b)$. Then we have

$$\int_a^b f(x)\,dx = 1 \implies c(b-a) = 1 \implies c = \frac{1}{b-a}.$$

Thus the probability density function of a continuous uniform distribution is

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}.$$

Notation: $X \sim U(a, b)$ means that X is a continuous uniform random variable on the interval (a, b).

CDF: The CDF of $X \sim U(a, b)$ is

$$F(x) = P(X \leq x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \geq b \end{cases}.$$

Mean and Variance: If $X \sim U(a, b)$, then

$$\mu = E(X) = \frac{a+b}{2}, \quad \sigma^2 = Var(X) = \frac{(b-a)^2}{12}.$$

Example: Suppose you arrie at a bus stop at 10:00 am. The bus will arrive at a time T uniformly distributed between 10:00 am and 10:30 am.
a. What is the probability that you will have to wait more than 5 minutes?
b. If at 10:15 am the bus has not arrived, what is the probability that you will have to wait at least another 5 minutes?
c. What is the probability that the bus will arrive exactly at 10:15 am?

a.

$$X \sim U(0, 30)$$
$$P(X > 5) = 1 - P(X \leq 5) = 1 - F(5) = 1 - \frac{5-0}{30-0} = \frac{25}{30} = \frac{5}{6}$$

b.

$$X \sim U(0, 30)$$
$$P(X > 20|X > 15) = \frac{P(X > 20 \cap X > 15)}{P(X > 15)} = \frac{P(X > 20)}{P(X > 15)} = \frac{1 - F(20)}{1 - F(15)}$$
$$= \frac{1 - \frac{20-0}{30-0}}{1 - \frac{15-0}{30-0}} = \frac{\frac{10}{30}}{\frac{15}{30}} = \frac{2}{3}$$

c.

$$P(X = 15) = \int_{15}^{15} f(x)\, dx = 0.$$

This is because the probability of a continuous random variable taking on any single value is 0.

4.2 Cumulative Distribution Function (CDF)

Definition: The cumulative distribution function (CDF) of a continuous random variable X is a function $F(x)$ defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

where $f(t)$ is the PDF of X.

Relationship between PDF and CDF:

- The CDF $F(x)$ is the integral of the PDF $f(x)$.
- The PDF $f(x)$ is the derivative of the CDF $F(x)$, i.e., $f(x) = \frac{d}{dx}F(x)$.

Properties of CDF:

- $F(x)$ is non-decreasing.
- $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.
- $P(X \leq x) = F(x)$
- $P(X > x) = 1 - F(x)$
- $P(a < X \leq b) = F(b) - F(a)$

Example: Let X be a continuous random variable with PDF:

$$f(x) = \begin{cases} \frac{2}{9}x & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

a. Find the CDF $F(x)$.
b. Use the CDF to compute $P(X > 2)$.

a.

$$F(x) = \int_0^x f(t)dt = \int_0^x \frac{2}{9}tdt = \begin{cases} 0 & x < 0 \\ \frac{2}{9} \cdot \frac{x^2}{2} = \frac{x^2}{9} & 0 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

b.

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) = 1 - F(2) \\ &= 1 - \frac{2^2}{9} = 1 - \frac{4}{9} = \frac{5}{9} \end{aligned}$$

Percentiles

Definition: The score s is the p-th percentile $\eta(p)$ of a continuous random variable X if:

- $P(X \leq s) = F(s) = p$
- $P(X > s) = 1 - F(s) = 1 - p$
- $\int_{-\infty}^{\eta(p)} f(x)dx = p$
- $F(\eta(p)) = p$

Percentile Vocabulary:

- $\eta(0.5)$ is the median.
- $\eta(0.25)$ is the first quartile Q_1.
- $\eta(0.75)$ is the third quartile Q_3.
- The interquartile range (IQR) is $Q_3 - Q_1$.

Example: Find the 75th percentile of the continuous random variable X from the previous example.

$$\begin{aligned}
\eta(0.75) &= s \text{ such that } F(s) = 0.75 \\
F(s) &= \frac{s^2}{9} = 0.75 \\
s^2 &= 6.75 \\
s &= \sqrt{6.75} \approx 2.598
\end{aligned}$$

Mean and Variance of a Continuous Random Variable

Mean (Expected Value): The mean or expected value of a continuous random variable X with PDF $f(x)$ is given by:

$$\begin{aligned}
E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\
E(h(X)) &= \int_{-\infty}^{\infty} h(x) f(x) dx \\
E(aX + b) &= aE(X) + b \\
E(h(X)) &\neq h(E(X)) \text{ unless } h(x) \text{ is linear}
\end{aligned}$$

Variance: The variance of a continuous random variable X with PDF $f(x)$ is given by:

$$\begin{aligned}
\sigma^2 = Var(X) &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \\
Var(X) &= E(X^2) - (E(X))^2 \\
Var(aX + b) &= a^2 Var(X)
\end{aligned}$$

Example: Let X be a continuous random variable with PDF:

$$f(x) = \begin{cases} \frac{3}{8}x^2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

a. Find $E(X)$
b. Find $E(X^2)$
c. Find $Var(X)$

a.

$$E(X) = \int_0^2 x \cdot \frac{3}{8}x^2 dx = \int_0^2 \frac{3}{8}x^3 dx = \left[\frac{3}{8} \cdot \frac{x^4}{4}\right]_0^2 = \frac{3}{8} \cdot 4 = \frac{3}{2}$$

b.

$$E(X^2) = \int_0^2 x^2 \cdot \frac{3}{8}x^2 dx = \int_0^2 \frac{3}{8}x^4 dx = \left[\frac{3}{8} \cdot \frac{x^5}{5}\right]_0^2 = \frac{3}{8} \cdot \frac{32}{5} = \frac{12}{5}$$

c.

$$Var(X) = E(X^2) - (E(X))^2 = \frac{12}{5} - \left(\frac{3}{2}\right)^2 = \frac{12}{5} - \frac{9}{4} = \frac{48}{20} - \frac{45}{20} = \frac{3}{20}$$

4.3 The Normal Distribution

The Normal Distribution is by far the most important distribution that we will study in this class. It is the distribution most often applied to data by scientists in all disciplines. It was first used in 733 by Moivre and used by Gauss to predict the location of astronomical bodies.

Also known as:

- Normal Distribution
- Gaussian Distribution
- Bell Curve

Definition A continuous random variable X is said to have a normal distribution with mean μ and variance σ^2 if its PDF is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad -\infty < x < \infty.$$

We denote this as $X \sim \mathcal{N}(\mu, \sigma^2)$.

Expected Value and Variance If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mu = E(X), \quad \sigma^2 = Var(X).$$

Note that because of the symmetry of normal distributions, $\mu = \eta(0.5) =$ median.

Standard Normal Distribution

A normal random variable with mean 0 and variance 1 is called a standard normal random variable. We denote this as $Z \sim \mathcal{N}(0, 1)$ with CDF $\Phi(z)$.

Remarks

- The standard normal distribution does not often arise naturally, but it serves as a tool for computing probabilities involving the normal distribution.

- Tables of values of $\Phi(z)$ are used to find probabilities for normal random variables.

Standard Normal Table A standard normal table gives values of $\Phi(z)$ for various values of z. To use the table, we find the row corresponding to the first two digits of z and the column corresponding to the second decimal place of z. The value at the intersection gives $\Phi(z)$.

Normal Percentiles The kth percentile $\eta(k)$ of a normal distribution is the value z_k such that $\Phi(z_k) = k$. To find z_k, we look up the value k in the body of the standard normal table and find the corresponding z value from the row and column headers.

Tail Probabilities Later in this course we will frequently work with right tail areas of the standard normal distribution. Let z_a denote the value on the measurement axis, for which the area under the standard normal curve to the right of z_a is equal to a.

$$z_a = (1-a)100\text{-th percentile}$$

Standardization **Any** normal random variable X with mean μ and standard deviation σ can be transformed into a random variable Z having a standard normal distribution via the standardization transformation:

$$Z = \frac{X-\mu}{\sigma}.$$

This allows us to compute probabilities for any normal random variable using the standard normal table. Let $X \sim \mathcal{N}(\mu, \sigma^2)$, then for any a and b,

$$P(a \leq X \leq b) = P\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

Example Let X = height of a random female student entering college in inches, where $X \sim \mathcal{N}(\mu = 65, \sigma^2 = 25)$.

a. What is the probability that a randomly selected female student has a height between 55 and 72.5 inches?

$$\begin{aligned} P(55 \leq X \leq 72.5) &= P\left(\frac{55-65}{5} \leq Z \leq \frac{72.5-65}{5}\right) \\ &= P(-2 \leq Z \leq 1.5) \\ &= \Phi(1.5) - \Phi(-2) \\ &= 0.9332 - 0.0228 \\ &= 0.9104. \end{aligned}$$

b. What is the height of a student such that only 5% of the students are taller than her? (Find the 95th percentile.)

$$\begin{aligned} P(X > h) &= 0.05 \\ P\left(Z > \frac{h-65}{5}\right) &= 0.05 \\ P\left(Z < \frac{h-65}{5}\right) &= 0.95 \\ \frac{h-65}{5} &= \Phi^{-1}(0.95) \\ \frac{h-65}{5} &= 1.645 \\ h - 65 &= 8.225 \\ h &= 73.225. \end{aligned}$$

Example: LSAT scores have an approximately Normal distribution with mean $\mu = 150$ and standard deviation $\sigma = 10$.

a. If you have a score of 167, then you did better than what percent of the people taking the test with you?

$$\begin{aligned} P(X < 167) &= P\left(Z < \frac{167 - 150}{10}\right) \\ &= P(Z < 1.7) \\ &= \Phi(1.7) \\ &\approx 0.9554. \end{aligned}$$

b. How high would your score have to be so that you are in the top 90%?

$$\begin{aligned} P(X > s) &= 0.10 \\ P\left(Z > \frac{s - 150}{10}\right) &= 0.10 \\ P\left(Z < \frac{s - 150}{10}\right) &= 0.90 \\ \frac{s - 150}{10} &= \Phi^{-1}(0.90) \\ \frac{s - 150}{10} &\approx 1.28 \\ s - 150 &\approx 12.8 \\ s &\approx 162.8. J \end{aligned}$$

4.4 The Exponential and Gamma Distributions

The Exponential Distribution

A continuous random variable X is said to have an Exponential distribution with parameter $\lambda > 0$ if its probability density function is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}.$$

Notation: $X \sim \text{Exp}(\lambda)$ means that X is an Exponential random variable with parameter λ.

CDF: The CDF of $X \sim \text{Exp}(\lambda)$ is

$$F(x) = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

Used for:

- The time until a radioactive particle decays.
- The time until a customer arrives at a service point.
- The time until a light bulb burns out.
- Life spans of living organisms.

Survival Function: The survival function is given by

$$S(t) = P(X > t) = 1 - F(t) = e^{-\lambda t}.$$

This is the probability that the event of interest has not occurred by time t. This is done instead of the CDF for lifespans because people are little bitches about death, so now we have to learn more formulae.

Mean and Variance: If $X \sim \text{Exp}(\lambda)$, then

$$\mu = E(X) = \frac{1}{\lambda}, \quad \sigma^2 = Var(X) = \frac{1}{\lambda^2}.$$

Note: When integrating by parts, let u be the polynomial term and dv be the exponential term.

Note: $\lim_{b\to\infty} \frac{\text{Polynomial}}{e^{\lambda b}} = 0$ for any polynomial.

Lack of Memory Property The Exponential distribution has the lack of memory property, which states that

$$P(X > s + t | X > s) = P(X > t) \forall s, t \geq 0.$$

This means that the probability of waiting an additional time t does not depend on how much time s has already elapsed.

Example: You are a customer service representative. Let X be the time in minutes that you spend with a customer. Suppose $X \sim \text{Exp}(\lambda = 0.5)$.

a. What is the probability that it takes the customer representitive more than 3 minutes with the next customer?

$$P(X > 3) = 1 - F(3) = e^{-0.5 \cdot 3} = e^{-1.5} \approx 0.2231.$$

b. Suppose the rep has already been on the phone for 15 minutes. What is the probability it will take a total of 18 minutes or more with this customer?

$$\begin{aligned} P(X > 18 | X > 15) &= \frac{P(X > 18 \cap X > 15)}{P(X > 15)} = \frac{P(X > 18)}{P(X > 15)} \\ &= \frac{e^{-0.5 \cdot 18}}{e^{-0.5 \cdot 15}} = e^{-0.5 \cdot 3} = e^{-1.5} \approx 0.2231. \end{aligned}$$

c. What is the probability that it takes the customer rep less than 3 minutes with the next5 customers?

$$
\begin{aligned}
&X_i \sim \text{Exp}(0.5) \quad i = 1, 2, 3, 4, 5 \\
P(X_1 < 3 \cap X_2 < 3 \cap X_3 < 3 \cap X_4 < 3 \cap X_5 < 3) &= P(X_1 < 3)P(X_2 < 3)P(X_3 < 3)P(X_4 < 3)P(\\
&= (F(3))^5 = (1 - e^{-0.5 \cdot 3})^5 \approx 0.6321^5 \approx 0.1013.
\end{aligned}
$$

d. What is the probability that it takes the rep a total of 15 minutes or less to help the next 5 customers?

$$X_i \sim \text{Exp}(0.5) \quad i = 1, 2, 3, 4, 5$$

$$T = \sum_{i=1}^{5} X_i \sim \Gamma(n = 5, \lambda = 0.5)$$

$$
\begin{aligned}
P(T \leq 15) = F_T(15) &= 1 - e^{-0.5 \cdot 15} \sum_{k=0}^{4} \frac{(0.5 \cdot 15)^k}{k!} \\
&= 1 - e^{-7.5} \left(1 + 7.5 + \frac{7.5^2}{2} + \frac{7.5^3}{6} + \frac{7.5^4}{24}\right) \approx 0.5596.
\end{aligned}
$$

The Gamma Distribution

Definition A continuous random variable with PDF

$$f(x) = \begin{cases} \frac{\lambda(\lambda x)^{n-1} e^{-\lambda x}}{(n-1)!} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

is said to have a Gamma distribution with parameters n and λ, where n is a positive integer and $\lambda > 0$.

The Gamma Function The Gamma function is defined as

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} \, dx \quad \alpha > 0.$$

For positive integers n, $\Gamma(n) = (n-1)!$.

Properties of the Gamma Function:

- For $\alpha > 1, \Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$.
- For a positive integer $n, \Gamma(n) = (n - 1)!$.
- $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Examples of Gamma Distribution

- Time until the r-th customer gets their coffee
- Milage that you get out of r tanks of gas - note that r does not have to be an integer here

Relationship between Gamma and Exponential Distributions Much like the negative binomial gets us the probability of the r-th success in a series of Bernoulli trials, the Gamma distribution gives us the probability of the r-th event occurring in an exponential process.

If $X_1, X_2, \ldots, X_r$ are independent Exponential random variables with parameter λ, then

$$T = \sum_{i=1}^{r} X_i \sim \Gamma(r, \lambda).$$

Mean and Variance: If $X \sim \Gamma(r, \lambda)$, then

$$\mu = E(X) = \frac{r}{\lambda}, \quad \sigma^2 = Var(X) = \frac{r}{\lambda^2}.$$

Note that the textbook defines the following:

$$\alpha = r, \quad \beta = \frac{1}{\lambda}.$$

The CDF of the Gamma Distribution For and arbitrary $r > 0$ there is no closed form for the CDF, but for positive integers r we have

$$F(x) = P(X \leq x) = 1 - \sum_{k=0}^{r-1} \frac{(\lambda x)^k e^{-\lambda x}}{k!}.$$

5. Joint Probability Distributions and Random Samples

5.1 Jointly Distributed Random Variables

Definition Two random variables X and Y are said to be **jointly distributed** if their outcomes are determined by the same random experiment. The **joint probability distribution** of X and Y gives the probabilities associated with all possible pairs of outcomes of X and Y. The pmf or pdf of the joint distribution is denoted by

$$P(x, y) = P(X = x, Y = y)$$

Example Suppose you have a hat with 3 balls labeled 1, 2, and 3. You randomly select one ball, record its number as X, then flip a coin that many times and record the number of heads as Y. The possible outcomes of this experiment are:

a. Find $P(1, 0)$:

$$P(1, 0) = P(X = 1, Y = 0) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

b. Find $P(1, 1)$:

$$P(1, 1) = P(X = 1, Y = 1) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

c. Find $P(2, 0)$:

$$P(2, 0) = P(X = 2, Y = 0) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

d. Find $P(2, 1)$:

$$P(2, 1) = P(X = 2, Y = 1) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

e. Find $P(2, 2)$:

$$P(2, 2) = P(X = 2, Y = 2) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

f. Find $P(3,0)$:

$$P(3,0) = P(X = 3, Y = 0) = \frac{1}{3} \cdot \frac{1}{8} = \frac{1}{24}$$

g. Find $P(3,1)$:

$$P(3,1) = P(X = 3, Y = 1) = \frac{1}{3} \cdot \frac{3}{8} = \frac{1}{8}$$

h. Find $P(3,2)$:

$$P(3,2) = P(X = 3, Y = 2) = \frac{1}{3} \cdot \frac{3}{8} = \frac{1}{8}$$

i. Find $P(3,3)$:

$$P(3,3) = P(X = 3, Y = 3) = \frac{1}{3} \cdot \frac{1}{8} = \frac{1}{24}$$

Marginal PMFs

The **marginal pmf** of a random variable is the pmf of that variable without reference to the other variable. We denote the marginal pmfs of X and Y as $P_X(x)$ and $P_Y(y)$, respectively. They can be found by summing over the joint pmf:

$$P_X(x) = \sum_y P(x,y)$$

$$P_Y(y) = \sum_x P(x,y)$$

Example (continued) Find the marginal pmfs of X and Y from the previous example.

$$P_X(1) = P(1,0) + P(1,1) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$
$$P_X(2) = P(2,0) + P(2,1) + P(2,2) = \frac{1}{12} + \frac{1}{6} + \frac{1}{12} = \frac{1}{3}$$
$$P_X(3) = P(3,0) + P(3,1) + P(3,2) + P(3,3) = \frac{1}{24} + \frac{1}{8} + \frac{1}{8} + \frac{1}{24} = \frac{1}{3}$$
$$P_Y(0) = P(1,0) + P(2,0) + P(3,0) = \frac{1}{6} + \frac{1}{12} + \frac{1}{24} = \frac{7}{24}$$
$$P_Y(1) = P(1,1) + P(2,1) + P(3,1) = \frac{1}{6} + \frac{1}{6} + \frac{1}{8} = \frac{11}{24}$$
$$P_Y(2) = P(2,2) + P(3,2) = \frac{1}{12} + \frac{1}{8} = \frac{5}{24}$$
$$P_Y(3) = P(3,3) = \frac{1}{24}$$

Trick For any X, Y,

$$P(X = Y) + P(Y > X) + P(X > Y) = 1$$

Independence

Two discrete random variables X and Y are said to be **independent** if for all values of x and y,

$$\forall (x,y) \in \mathbb{R}, \quad P(X = x, Y = y) = P_X(x) \cdot P_Y(y)$$

Conditional Distributions

Definition The **conditional pmf** of Y given $X = x$ is defined as

$$P_{Y|X}(y|x) = P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P_X(x)}$$

Similarly, the conditional pmf of X given $Y = y$ is defined as

$$P_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P_Y(y)}$$

Example (continued) :

a. Find $P_{X|Y}(3|y = 2)$:

$$P_{X|Y}(3|2) = \frac{P(3,2)}{P_Y(2)} = \frac{\frac{1}{8}}{\frac{5}{24}} = \frac{3}{5}$$

b. Find $P_{Y|X}(3|x = 2)$:

$$P_{Y|X}(3|2) = \frac{P(2,3)}{P_X(2)} = \frac{0}{\frac{1}{3}} = 0$$

c. Find $P_{Y|X}(1|x = 3)$:

$$P_{Y|X}(1|3) = \frac{P(3,1)}{P_X(3)} = \frac{\frac{1}{8}}{\frac{1}{3}} = \frac{3}{8}$$

d. Find conditional pmf of X given $Y = 1$:

$$P_{X|Y}(1|1) = \frac{P(1,1)}{P_Y(1)} = \frac{\frac{1}{6}}{\frac{11}{24}} = \frac{4}{11}$$
$$P_{X|Y}(2|1) = \frac{P(2,1)}{P_Y(1)} = \frac{\frac{1}{6}}{\frac{11}{24}} = \frac{4}{11}$$
$$P_{X|Y}(3|1) = \frac{P(3,1)}{P_Y(1)} = \frac{\frac{1}{8}}{\frac{11}{24}} = \frac{3}{11}$$

5.2 Expected Values, Covariance, and Correlation

Definition Let X and Y be jointly distributed random variables with pmf or pdf $P(x,y)$. The **expected value** of a function $h(X,Y)$ is given by

$$E[h(X,Y)] = \sum_x \sum_y h(x,y)P(x,y) \quad \text{(discrete)}$$

or

$$E[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) f(x,y)\, dx\, dy \quad \text{(continuous)}$$

Expected values and variances can be used to describe the behavior of a single random variable. If we want to consider joint behavior, we need to consider how one quantity influence the other?

Covariance

Definition The **covariance** between two random variables X and Y is defined as

$$Cov(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - \mu_X\mu_Y$$

In practice, the following computational formula is typically used:

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

The covarience describes the joint behavior of the variables X and Y. However, this quantity depends on the measurement units of X and Y.

Correlation Coefficient

The correlation coefficient of X and Y is denoted by $\rho_{X,Y}$ and is defined as

$$\rho_{X,Y} = \frac{Cov(X,Y)}{\sigma_X\sigma_Y}$$

The correlation coefficient is a unitless measure of the linear relationship between X and Y. Its value is always between -1 and 1. A value of 1 indicates a perfect positive linear relationship, a value of -1 indicates a perfect negative linear relationship, and a value of 0 indicates no linear relationship.

Properties of Covariance and Correlation

- $Cov(X, X) = Var(X)$
- $Cov(X, Y) = Cov(Y, X)$
- $Cov(aX, bY) = ab\,Cov(X, Y)$
- $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$
- If X and Y are independent, then $Cov(X, Y) = 0$ and $\rho_{X,Y} = 0$
- $\rho = \pm 1 \iff \exists a, b \in \mathbb{R} \ni Y = aX + b$

5.3 Random Samples and Statistics

Definition The RBs $X_1, X_2, \ldots, X_n$ are said to form a **random sample** of size n If

- The X_i are independent
- Every X_i has the same probability distribution

Any function of a random sample $Y = T(X_1, X_2, \ldots, X_n)$ is called a **statistic**.

Some Common Statistics

- Sample Mean: $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- Sample Total: $T = \sum_{i=1}^{n} X_i$
- Sample Variance: $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$

Sampling Distributions

Definition The probability distribution of a statistic is called a **sampling distribution**. In this case we are especially interested in the sampling distributions of the sample mean $\bar{X}$ and the sample variance S^2. The standard deviation of a sampling distribution is called the **standard error** of the statistic.

Proposition Let $X_1, X_2, \ldots, X_n$ be a random sample from some distribution. This means that the observations are independent and have the same distribution. Let $\bar{X}$ be the sample mean. Then

- $E[\bar{X}] = \mu$
- $Var(\bar{X}) = \frac{\sigma^2}{n}$
- $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

When the Population is Normal

Proposition Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution $\mathcal{N}(\mu, \sigma^2)$. Then the sample mean $\bar{X}$ is also normally distributed with distribution

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

In Summary

- The sample mean $\bar{X}$ is a random variable
- Therefore it has a probability distribution
- If the population is also normal, then the sample mean with also be normal
- A normal distribution is described by its mean and variance
- If we sampled from a distribution with mean μ and variance σ^2, then the sample mean will also have mean μ, but have variance $\frac{\sigma^2}{n}$

Example: Let $X_1 \ldots X_{25} \sim \mathcal{N}(106, 144)$.

a. What can you say about the sampling distribution of $\bar{X}$?

$$\bar{X} \sim \mathcal{N}\left(106, \frac{144}{25}\right) = \mathcal{N}(106, 5.76)$$

b. Find $P(X > 110)$

$$\begin{aligned} P(X > 110) &= P\left(Z > \frac{110 - 106}{\sqrt{144/25}}\right) = P(Z > 1.67) \\ &= 1 - P(Z < 1.67) = 1 - 0.9525 = 0.0475 \end{aligned}$$

c. Find $P(\bar{X} > 110)$

$$
\begin{aligned}
P(100 < \bar{X} < 110) &= P\left(\frac{100-106}{\sqrt{144/25}} < Z < \frac{110-106}{\sqrt{144/25}}\right) \\
&= P(-2.5 < Z < 1.67) = P(Z < 1.67) - P(Z < -2.5) \\
&= 0.9525 - 0.0062 = 0.9463
\end{aligned}
$$

5.4 Distribution of Linear Combinations of Random Variables

Definition Given a collection of random variables $X_1, X_2, \ldots, X_n$, a **linear combination** of these random variables is a new random variable Y defined as

$$Y = a_1X_1 + a_2X_2 + \ldots + a_nX_n = \sum_{i=1}^{n} a_iX_i$$

where $a_1, a_2, \ldots, a_n$ are constants.

Mean and Variance of Linear Combinations If $X_1, X_2, \ldots, X_n$ are random variables with means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, then the mean and variance of the linear combination Y are given by:

$$E[Y] = \sum_{i=1}^{n} a_iE[X_i] = \sum_{i=1}^{n} a_i\mu_i$$

regardless of whether the random variables are independent or not, and

$$Var(Y) = \sum_{i=1}^{n} a_i^2 Var(X_i) = \sum_{i=1}^{n} a_i^2\sigma_i^2$$

if the random variables $X_1, X_2, \ldots, X_n$ are independent, or

$$Var(Y) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_jCov(X_i, X_j)$$

Properties of Covariance

- $Cov(X, X) = Var(X)$
- $Cov(X, Y) = Cov(Y, X)$
- $Cov(aX, bY) = ab\,Cov(X, Y)$
- $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$
- If X and Y are independent, then $Cov(X, Y) = 0$

Example: Let X_1 and X_2 be random variables with means $\mu_1 = 1$ and $\mu_2 = 5$, variances $\sigma_1^2 = 1$ and $\sigma_2^2 = 4$, and covariance $Cov(X_1, X_2) = 3$.

a. Are X_1 and X_2 independent? No (since $Cov(X_1, X_2) \neq 0$)
b. Find $E(X_1 + 2X_2)$

$$E(X_1 + 2X_2) = E(X_1) + 2E(X_2) = 1 + 2(5) = 11$$

c. Find $Var(X_1 + 2X_2)$

$$Var(X_1+2X_2) = Var(X_1)+4Var(X_2)+4Cov(X_1, X_2) = 1+4(4)+4(3) = 29$$

d. Find $V(X_1 - 2X_2)$

$$Var(X_1-2X_2) = Var(X_1)+4Var(X_2)-4Cov(X_1, X_2) = 1+4(4)-4(3) = 5$$

6. Point Estimation

6.1 Point Estimation

Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with mean μ and finite variance σ^2. Then, as n approaches infinity, the distribution of the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ approaches a normal distribution with mean μ and variance $\frac{\sigma^2}{n}$, regardless of the shape of the original population distribution. Formally,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \text{ as } n \to \infty.$$

Usually $n \geq 30$ is considered sufficient for the CLT to hold.

Point Estimation

Let θ be an unknown parameter of a population distribution, and let $X_1, X_2, \ldots, X_n$ be a random sample from that distribution. An estimator $\hat{\theta} = g(X_1, X_2, \ldots, X_n)$ is a function of the sample data used to estimate the parameter θ. The estimator $\hat{\theta}$ is called a **point estimator** of the parameter θ.

If $E(\hat{\theta}) = \theta$, then $\hat{\theta}$ is an **unbiased estimator** of θ. $\sigma_{\hat{\theta}}$ is called the **standard error** of the estimator $\hat{\theta}$.

We will study 2 cases:

Case 1: Quantitative variable Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with mean μ and variance σ^2. The parameter we want to target is $\mu = E(X_i)$. In this case, $\hat{\theta} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Case 2: Binary Variable Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with proportion p. The parameter we want to target is P :

$$\begin{cases} \text{Proportion of indifividuals with characteristic} & \text{if } X_i \sim \text{Bernoulli}(p) \\ \text{Probability of success} & \text{if } X_i \sim \text{Binomial}(n,p) \end{cases}$$

Parameter	Point Estimator	Bias?	Standard Error	Estimated Standard Error
μ	$\bar{X} = \frac{1}{n}\sum X_i$	$\frac{sigma}{\sqrt{n}}$	$\frac{S}{\sqrt{n}}$	
p	$\hat{p}$: sample proportion	$E(\hat{p}) = p$	$\sqrt{\frac{p(1-p)}{n}}$	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Example: Let X_i be the weight of banana i and μ_X be the average weight of a banana. We take a sample of n bananas by weighing them. We will use $\mu_X = \frac{1}{n}\sum X_i$ as a point estimator.

$$E(\bar{X}) = \mu, \text{so } \bar{X} \text{ is an unbiased estimator of } \mu.$$

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{\sigma}{\sqrt{n}}$$

Example: Parameter of interest is p = proportion of California Residents with an active library card. Point estimator $\hat{p}$ = # in your sample of CA Residents with an active library card.

$$X_i = \begin{cases} 1 & \text{if individual } i \text{ has an active library card} \\ 0 & \text{otherwise} \end{cases} \quad \hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$T = \sum_{i=1}^{n} X_i \sim \mathrm{Binomial}(n, p)$$

$$E(\hat{p}) = E\left(\frac{T}{n}\right) = \frac{1}{n}E(T) = \frac{1}{n}(np) = p$$

$$\mathrm{Var}(\hat{p}) = \mathrm{Var}\left(\frac{T}{n}\right) = \frac{1}{n^2}\mathrm{Var}(T) = \frac{1}{n^2}(np(1-p)) = \frac{p(1-p)}{n}$$

Properties of $\hat{\theta}$:

- $E(\hat{\theta}) = E(\frac{T}{n}) = \frac{1}{n}E(T) = \theta$ so $\hat{\theta}$ is an unbiased estimator of θ.
- $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}(\frac{1}{n}T) = \frac{1}{n^2}\mathrm{Var}(T) = \frac{1}{n^2}(n\theta(1-\theta)) = \frac{\theta(1-\theta)}{n}$

Note: $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is an unbiased estimator of σ^2.

Standard Error

Definition The **standard error** of an estimator $\hat{\theta}$ is it's standard deviation $\sigma_{\hat{\theta}} = \sqrt{Var(\hat{\theta})}$

Sometimes, the standard error itself involves unknown parameters. In this case, the standard error can be estimated from the sample. The estimate of the standard error $\sigma_{\hat{\theta}}$ is called the **estimated standard error** and is denoted $s_{\hat{\theta}}$

Recall the probability distribution of a statistic is called a **sampling distribution**.

Sampling Distribution of the Sample Mean

Proposition: Let $X_1 \dots X_n$ be a random sample from soe distribution. This means that the observations $X_1 \dots X_n$ are independent and have the same distribution (same μ and σ^2). Let $\bar{X}$ denote the cample mean, then

- $E(\bar{X}) = \mu$
- $\sigma^2_{\bar{X}} = \frac{\sigma^2}{n}$
- $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

What is the standard error of $\bar{X}$?

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \text{ (standard error of the sample mean)}$$

Point Estimates and Probability

Usually, point estimates are chosen because we hope that their values for random samples are close to the true (unkown) population paramets. But how close? If we know the distribution of $\bar{X}$, we can compute the probability that $\bar{X}$ will fall into any specific interval. In particular, you can compute the probability that $\bar{X}$ will be no further than a constant c from the true mean μ:

General Form Assume that $X_1, \ldots, X_n$ is a random sample from a Normal distribution with an unknown mean and a known variance σ^2. Then $\bar{X}$ has a Normal distribution with mean μ and variance $\frac{\sigma^2}{n}$. Hence,

$$P(|\bar{X} - \mu| \leq c) = P\left(-c \leq \bar{X} - \mu \leq c\right) = P\left(\frac{-c}{\sigma/\sqrt{n}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{c}{\sigma/\sqrt{n}}\right)$$

Using the standard normal distribution, we get

$$P(|\bar{X} - \mu| \leq c) = P\left(Z \leq \frac{c\sqrt{n}}{\sigma}\right) - P\left(Z \leq \frac{-c\sqrt{n}}{\sigma}\right) = 2P\left(Z \leq \frac{c\sqrt{n}}{\sigma}\right) - 1$$

Confidence Interval

Main Idea: If $\bar{X}$ is close to μ with 95% probability, then μ is also close to $\bar{X}$ with 95% probability. Using this idea, we can use our point estimate $\bar{X}$ to create a **confidence interval** for μ.

Definition: After observing a random sample from a normal population of size n and computing the sample mean $\bar{X}$, a 95% confidence interval for μ itself

$$CI_\mu = (\bar{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}})$$

This means that with 95% probability, the interval CI_μ contains the true mean μ.

Example: A random sample of $n = 50$ males showed an average daily intake of dairy products equal to 756 grams. Suppose it is known that dairy intake follows a normal distribution with standard deviation of 35 grams. Fins a 95% confidence interval for the population mean μ.

$$CI_\mu = (\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}) = (756 - 1.96 \cdot \frac{35}{\sqrt{50}}, 756 + 1.96 \cdot \frac{35}{\sqrt{50}}) = (744.26, 767.74)$$

Interpreting Confidence Intervals: **Warning:** Assume that you collect data $(X_1, \ldots, X_n)$ and compute a 95% confidence interval. It is tempting, but wrong, to say that your confidence interval contains the true mean μ with 95% probability. Since the true mean is a fixed number, it will be contained in any interval with probability either 0 or 1. The **Confidence Level** (here 95%) rather refers to the generation of the random variables or the sampling process.

Confidence Levels: To compute conficence intervals for levels other than 95%, we have to change the critical z-value from 1.96 to another number. You can find the appropriate number in the Normal CDF table. Therefore, the general formula for a confidence interval with confidence level $100(1-\alpha)\%$ is:

$$CL_{\mu} = (\bar{X} - z_{\frac{\sigma}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\sigma}{2}} \frac{\sigma}{\sqrt{n}})$$

Confidence Level, Precision, and Sample Size: Which is better? A 95% confidence interval or a 99% confidence interval? It depends on how much precision is required. Higher confidence levels lead to wider intervals, therefore less precision. If more precision is required, a larger sample size is needed. The following factors influence the width of a confidence interval:

- Confidence Level: Higher confidence levels lead to wider intervals.
- Sample size: Larger sample sizes lead to narrower intervals.
- Standard deviation of the population: Larger standard deviations lead to wider intervals.

Confidence Intervals for Large Samples: If the sample size n is large then the sample mean has approximately a Normal distribution regardless of the population distribution according to teh Cenreal Limit Theorem. However, in most practical applications the population variance σ^2 is unknown. It is still possible to obtain conficence intervals for the mean, by first estimating σ^2 by s^2 and then using the estimate s in the conficence interval formula. If n is sufficiently large (usually $n \geq 40$), then the standardized variable

$$Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim \mathcal{N}(0,1)$$

has approximately a standard normal distribution. The n needs to be a little bit larger here because of the additional variation introduced through the estimation of σ.

Definition: After observing a large ($n \geq 40$) random sample from a population with unknown variance σ^2 and computing the sample mean $\bar{X}$, an approximate $100(1-\alpha)\%$ confidence interval for μ is given by

$$CI_\mu = (\bar{X} - z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}})$$

Example: Suppose the instructor of a large calculus class tries to write an exam of medium difficulty with a mean score of 75. Fifty students take the exam. The mean of these exam scores is 82 with a standard deviation of 18.

a. Compute a 90% confidence interval for the degree of difficulty of the exam.

$X =$ a student's test score $\quad \bar{X} = 75 \quad n = 50 \quad \mu_X = 82 \quad \sigma_X = 18 \quad \alpha = .1$

$$\begin{aligned}
CI_\mu &= (\bar{X} - z_0.05\frac{\sigma}{\sqrt{n}}, \bar{X} + z_0.05\frac{\sigma}{\sqrt{n}}) \\
&= (75 + 1.64\frac{18}{\sqrt{50}}, 75 - 1.64\frac{18}{\sqrt{50}}) \\
&= (75 + 4.175, 75 - 4.175) = (79.175, 70.825) \\
&= (70.825, 79.175)
\end{aligned}$$

b. In light of this information, we can conclude the exam turned out slightly easier than the professor had hoped.